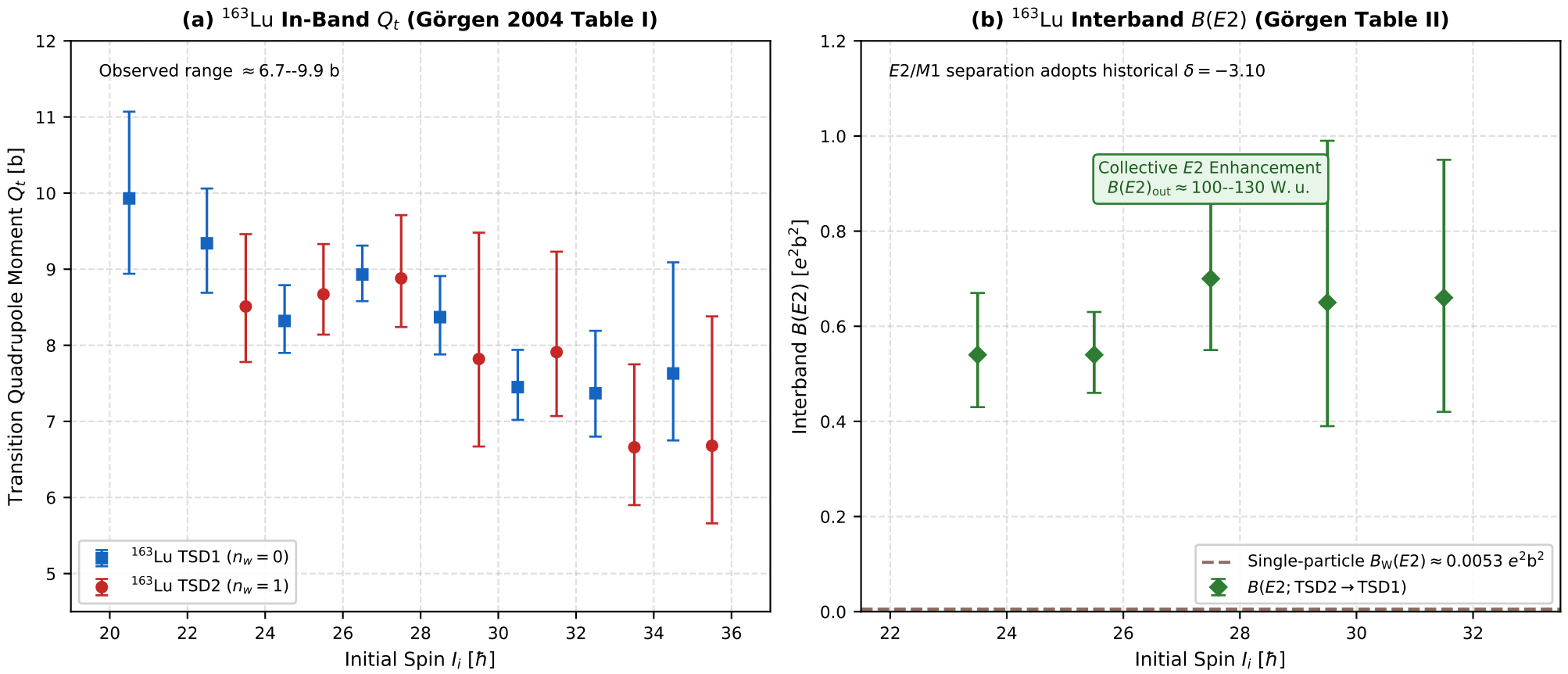


(c) Orientation Modes & Alignment Geometry

Transverse Wobbling (TW): ^{163}Lu, ^{165,167}Lu TSD

Model geometry: **j** mainly on *s*; maximum inertia on *m*; **R** need not be exactly on *m*

High core spin $R \sim 15$--$30\hbar \gg 1$; Strong collective $\Delta I = 1$ $E2$ linking transitions

High-K Precession: ^{178}W, ^{178,179}Hf (Axial Limit)

Axially symmetric limit ($\gamma \to 0$); Single-particle alignments build K on symmetry axis

Axial shape constraint $R_3 = 0$; fixed-K rotational band remains; first gap $\hbar^2(K+1)/\mathcal{J}_\perp$

Competing interpretations: ^{135}Pr (Lv 2022), ^{187}Au (Guo 2022)

Small-$|\delta|$ solutions (747 keV: $R_{ac} = 0.37$, $\delta = -0.47$); $M1$ fractions derived from δ

Single-particle re-alignment towards rotation axis / γ-soft IBFM / shape coexistence